Abstract

We study the topological properties of phonons in trapped Rydberg atom arrays, which arise from dipole-dipole interactions between the atoms. For various one-dimensional geometries, from zigzag to armchair configurations, we analyze the symmetries of the phononic Hamiltonian which give rise to topologically localized phonon excitations on the system boundaries. Because the phonon-phonon interactions here do not naturally respect chiral symmetry, which is crucial for realizing 1D topological phases, we show how an appropriate, geometry-dependent choice of the dipole moment orientation can restore this symmetry and enable topological characterization. The interplay between two phononic degrees of freedom in the harmonic trap potential, mediated by the dipole-dipole interactions, leads to interesting topological phases with winding numbers between zero and two. The corresponding edge states exhibit strong localization and robustness, making them potentially useful for applications in Rydberg-array-based quantum transport.

As an outlook we aim to explore how the phonon topology influences transport phenomena. Topological states in the electronic degrees of freedom in Rydberg arrays have already been shown to support topologically protected photon pumping [1–5]. Studying the effect of spin-phonon coupling can provide new control mechanisms in the realm of topological transport.

Hamiltonian

I am considering the interaction term H_I of a Hamiltonian of a system of N particles with positions $\{\bar{\mathbf{R}}_l\}_{l\in\{1,\ldots,N\}}$. It shall consist of two-body interactions, which depend on the position of the involved particles. Thus H_I can be written in the form

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j;\, i>j} V_{i,j}(\bar{\mathbf{R}}_i, \bar{\mathbf{R}}_j)$$

In order to being able to derive the second order expansion of H_I, that will be needed later, in a more formal way, I write the two-body interactions as dependend on all positions.

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j;\, i>j} V_{i,j}(\{\bar{\mathbf{R}}_l\})$$

The $V_{i,j}$ are formally functions of all positions $\{\bar{\mathbf{R}}_l\}$, i.e $V_{i,j}(\{\bar{\mathbf{R}}_l\})$. But of course they only depend on $\bar{\mathbf{R}}_i$ and $\bar{\mathbf{R}}_j$, i.e.

$$\frac{\partial V_{i,j}}{\partial \bar{\mathbf{R}}_l} = 0 \quad \text{if } i \neq l \neq j$$

Now I expand the interaction Hamiltonian in small displacements $\{\delta\mathbf{R}_l\}$ from the tweezer centers $\{\mathbf{R}_l\}$.

$$\begin{aligned}
H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n,m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m + \frac{1}{2}\sum_{n=m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\, i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m \\
&+ \frac{1}{2}\sum_n\sum_{i,j;\, i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n^2}\,\delta\mathbf{R}_n \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\, i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m\,(\delta_{n=i,m=j} + \delta_{n=j,m=i}) \\
&+ \frac{1}{2}\sum_n\sum_{i,j;\, i>j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n^2}\,\delta\mathbf{R}_n\,(\delta_{n=i} + \delta_{n=j})
\end{aligned}$$

Using the fact, that

$$\sum_{i,j;\, i>j} \left(\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \, \delta\mathbf{R}_i^2 + \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j^2} \, \delta\mathbf{R}_j^2 \right) = \sum_{i,j;\, i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \, \delta\mathbf{R}_i^2$$

and

$$\sum_{i,j;\, i>j} \left(\delta\mathbf{R}_i \, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \, \delta\mathbf{R}_j + \delta\mathbf{R}_j \, \frac{\partial^2 V_{j,i}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j \partial \mathbf{R}_i} \, \delta\mathbf{R}_i \right) = \sum_{i,j;\, i\neq j} \delta\mathbf{R}_i \, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \, \delta\mathbf{R}_j$$

and similar for the mixed terms,the expansion can be written as

$$\begin{aligned} H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \, \delta\mathbf{R}_i + \frac{1}{2} \sum_{i,j;\, i>j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \, (\delta\mathbf{R}_i \, \delta\mathbf{R}_j + \delta\mathbf{R}_j \, \delta\mathbf{R}_i) \\ &\quad + \frac{1}{2} \sum_{i,j;\, i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \, \delta\mathbf{R}_i^2 \\ &= H_I(\{\mathbf{R}_l\}) + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \, \delta\mathbf{R}_i + \frac{1}{2} \sum_{i,j;\, i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \, \delta\mathbf{R}_i \, \delta\mathbf{R}_j \\ &\quad + \frac{1}{2} \sum_{i,j;\, i\neq j} \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \, \delta\mathbf{R}_i^2 \end{aligned}$$

If I introduce a matrix U the expanded Hamiltonian can be brought into the same form as in the paper.

$$H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} \, \delta\mathbf{R}_n + \sum_{n,m} U_{n,m} \, \delta\mathbf{R}_n \, \delta\mathbf{R}_m$$

with

$$\begin{aligned} U_{n,m} &= \frac{1}{2} \, \frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \frac{1}{2} \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \end{aligned}$$

Swithing again to the reduced writting, where V_{ij} is only writen as a function of the variables of which it actually depends, reads

$$\begin{aligned} U_{n,m} &= \frac{1}{2} \, \frac{\partial^2 V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \frac{1}{2} \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{R}_n, \mathbf{R}_j)}{\partial \mathbf{R}_n^2} \end{aligned}$$

This result is a good point to further specify the potential that is used in the next sections. The potential will only depend on the relative vector between the two positions, i.e. $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$. Using the derivative

$$\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_i} = \mathbb{1} = -\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_j}$$

the derivatives of the U matrix can be written as

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} &= \frac{\partial}{\partial \mathbf{R}_n} \left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \, \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \right) \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \, \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \, \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \, \frac{\partial^2 \mathbf{r}_{nm}}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \\ &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \end{aligned}$$

and

$$
\begin{aligned}
\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_m^2 m} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \\
&= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}
\end{aligned}
$$

Thus

$$
\begin{aligned}
U_{n,m} &= -\frac{1}{2} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m \\
U_{n,n} &= \frac{1}{2} \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}
\end{aligned}
$$

At this point I can eliminate the linear term of the Hamiltonian by introducing the trap potential

$$
H_0(\{\delta \mathbf{R}_l\}) = \sum_i \frac{\mathbf{P}_i^2}{2m} + \frac{1}{2} m\omega^2 \, \delta \mathbf{R}_i^2,
$$

where $\mathbf{P}_i$ is the momentum of the ith atom. I can now shift the considered displacements from the trap centers by a constant $\{\delta \mathbf{R}_l\} = \{\delta \tilde{\mathbf{R}}_l + \boldsymbol{\Delta}_l\}$, where $\delta \mathbf{R}_l$ and $\delta \tilde{\mathbf{R}}_l$ are operators and $\boldsymbol{\Delta}_l$ are just constant vectors. Neglecting all constant terms the Hamiltonian minus the momentum term $\tilde{H} = H - \sum_i \mathbf{P}_i / 2m$ up to second order reads

$$
\begin{aligned}
\tilde{H}(\{\delta \mathbf{R}_l\}) &= \sum_i \frac{1}{2} m\omega^2 \, (\delta \tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i)^2 + \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} \, (\delta \tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i) + \sum_{i,j} (\delta \tilde{\mathbf{R}}_i + \boldsymbol{\Delta}_i) \, U_{i,j} \, (\delta \tilde{\mathbf{R}}_j + \boldsymbol{\Delta}_j) \\
&= \sum_i \frac{1}{2} m\omega^2 \, \delta \tilde{\mathbf{R}}_i^2 + \sum_{i,j} \delta \tilde{\mathbf{R}}_i \, U_{i,j} \, \delta \tilde{\mathbf{R}}_j + \sum_i \left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} + \sum_j \left(2\, U_{j,i} + m\omega^2 \, \delta_{ij} \right) \boldsymbol{\Delta}_j \right) \delta \tilde{\mathbf{R}}_i,
\end{aligned}
$$

where it was used that U is symmetric. As the matrix $\left(2\, U_{j,i} + m\omega^2 \, \delta_{ij} \right)$ is invertible, a set of vectors $\{\boldsymbol{\Delta}_n\}$ can be found such that the linear term vanishes. Thus

$$
H(\{\delta \tilde{\mathbf{R}}_l\}) = \sum_n \frac{\mathbf{P}_n^2}{2m} + \sum_n \frac{1}{2} m\omega^2 \, \delta \tilde{\mathbf{R}}_n^2 + \sum_{n,m} U_{n,m} \, \delta \tilde{\mathbf{R}}_n \, \delta \tilde{\mathbf{R}}_m
$$

At this point I introduce the quantized displacements in form oft the quadrature operators

$$
\begin{aligned}
\delta \tilde{\mathbf{R}}_n &= \sqrt{\frac{\hbar}{2m\omega}} \, (a_n + a_n^\dagger) \\
\mathbf{P}_n &= i \sqrt{\frac{\hbar m\omega}{2}} \, (a_n^\dagger - a_n)
\end{aligned}
$$

With that the Hamiltonian gets its final form

$$
H = \sum_n \hbar\omega \, a_n^\dagger a_n + \frac{\hbar}{2m\omega} \sum_{n,m} U_{n,m} \left(a_n \, a_m + a_n^\dagger a_m + a_n \, a_m^\dagger + a_n^\dagger a_m^\dagger \right)
$$

Physical implementation

As a physical implementation I consider a system of Rydberg atoms. The interaction stems from dipole-dipole interactions between mixed Rydberg states. The corresponding potential reads

$$
\begin{aligned}
V_{dd}(\mathbf{r}) &= \frac{\langle d\rangle^2}{4\pi\varepsilon_0\, r^3}\left(1-3\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2\right) =: V_0(r)\, f(\hat{\mathbf{r}}) = V_0(a)\,\frac{a^3}{r^3}\, f(\hat{\mathbf{r}})\\
\frac{\partial V_{dd}}{\partial r_\alpha} &= \frac{\langle d\rangle^2}{4\pi\varepsilon_0\, r^3}\,\frac{1}{r}\left(-3\,\frac{r_\alpha}{r}+15\,\frac{r_\alpha}{r}(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2-6\,\hat{d}_\alpha\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})\right) =: \frac{V_0(r)}{r}\, g(\hat{\mathbf{r}}) = \frac{V_0(a)}{a}\,\frac{a^4}{r^4}\, g(\hat{\mathbf{r}})\\
\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= \frac{\langle d\rangle^2}{4\pi\varepsilon_0\, r^3}\,\frac{1}{r^2}\left[\delta_{\alpha\beta}\left(-3+15\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2\right)-6\,\hat{d}_\alpha\hat{d}_\beta+30\,\frac{\hat{d}_\alpha r_\beta+\hat{d}_\beta r_\alpha}{r}\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})+15\,\frac{r_\alpha r_\beta}{r^2}\left(1-7\,(\hat{\mathbf{r}}\cdot\hat{\mathbf{d}})^2\right)\right]\\
&=: \frac{V_0(r)}{r^2}\, c(\hat{\mathbf{r}}) = \frac{V_0(a)}{a^2}\,\frac{a^5}{r^5}\, c(\hat{\mathbf{r}})
\end{aligned}
$$

The interaction strength $V_0(r) = \langle d\rangle^2/(4\pi\varepsilon_0\, r^3)$ can reach from a few MHz to a few GHz, when one assumes a dipole moment between $1000\, ea_0$ and $5000\, ea_0$ and a distance between the atoms of 3 to $10\,\mu$m ($a_0 = 4\pi\varepsilon_0\hbar^2/(e^2 m_e)$ is the Bohr radius). The trap frequency ω usually can take values of 10 to 100 kHz.

In the following I will consider a chain of atoms and will restrict the displacements of the atoms to the x direction, which is the direction in which the one dimensional chain extends. The chain consists of two sublattices A and B with a distance of $a\,\hat{\mathbf{e}}_x$ between neighboring atoms of the same sublattice and a distance of $b\,\hat{\mathbf{e}}_x + h\,\hat{\mathbf{e}}_y$ between neighboring atoms of different sublattices. Interactions shall only be relevant between nearest neighbors.

At this point let us rediscuss the shift of the displacements which allowed us to eliminate the first order term in the Hamiltonian. The necessary condition was

$$
\left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} + \sum_j \left(2\,U_{j,i} + m\omega^2\,\delta_{ij}\right)\,\Delta_j\right) = 0 \quad \forall i
$$

whereas

$$
\begin{aligned}
\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i} &= \sum_{n,m;\,n<m} \frac{\partial V_{n,m}(\mathbf{R}_n,\mathbf{R}_m)}{\partial \mathbf{R}_i}\,(\delta_{i=n}+\delta_{i=m})\\
&= \sum_{m;\,i<m} \frac{\partial V_{i,m}(\mathbf{R}_m-\mathbf{R}_i)}{\partial \mathbf{R}_i} + \sum_{n;\,n<i} \frac{\partial V_{n,i}(\mathbf{R}_i-\mathbf{R}_n)}{\partial \mathbf{R}_i}\\
&= \sum_{m;\,i<m} \frac{\partial V_{i,m}(\mathbf{R}_m-\mathbf{R}_i)}{\partial \mathbf{r}_{i,m}} - \sum_{n;\,n<i} \frac{\partial V_{n,i}(\mathbf{R}_i-\mathbf{R}_n)}{\partial \mathbf{r}_{n,i}}
\end{aligned}
$$

with $\mathbf{r}_{i,j} = \mathbf{R}_j - \mathbf{R}_i$. By defining

$$
\begin{aligned}
\tilde{g}_i &:= \frac{a}{V_0(a)}\,\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i}\\
\tilde{U}_{i,j} &:= \frac{a^2}{V_0(a)}\,2\,U_{j,i}
\end{aligned}
$$

The condition for the shift can be written as

$$
\frac{V_0(a)}{a}\,G_i := \frac{V_0(a)}{a}\left(\tilde{g}_i + \sum_j W_{ij}\,\frac{\Delta_j}{a}\right) := \frac{V_0(a)}{a}\left(\tilde{g}_i + \sum_j \left(\tilde{U}_{j,i} + \frac{m\omega^2 a^2}{V_0(a)}\,\delta_{ij}\right)\frac{\Delta_j}{a}\right) = 0 \quad \forall i
$$

Here h and $\tilde{U}$ are dimensionless and the the term $m\omega^2 a^2/[V_0(a)\max(\tilde{U})]$ is of order 10^{-15}-10^{-10}, hence negligibly small. Now I use the two free parameters h and the angle φ_d of the dipole moment $\hat{\mathbf{d}}$ to set $G_i = 0$ or close to zero for different values of b. Close to zero means here that $\max(|G_i|)$ compared to $\sqrt{\hbar/(2m\omega)}\max(\tilde{U})/a$ is small. Here $\sqrt{\hbar/(2m\omega)}/a \sim 10^{-3}$ is the displacement from a single motional excitation in units of a.

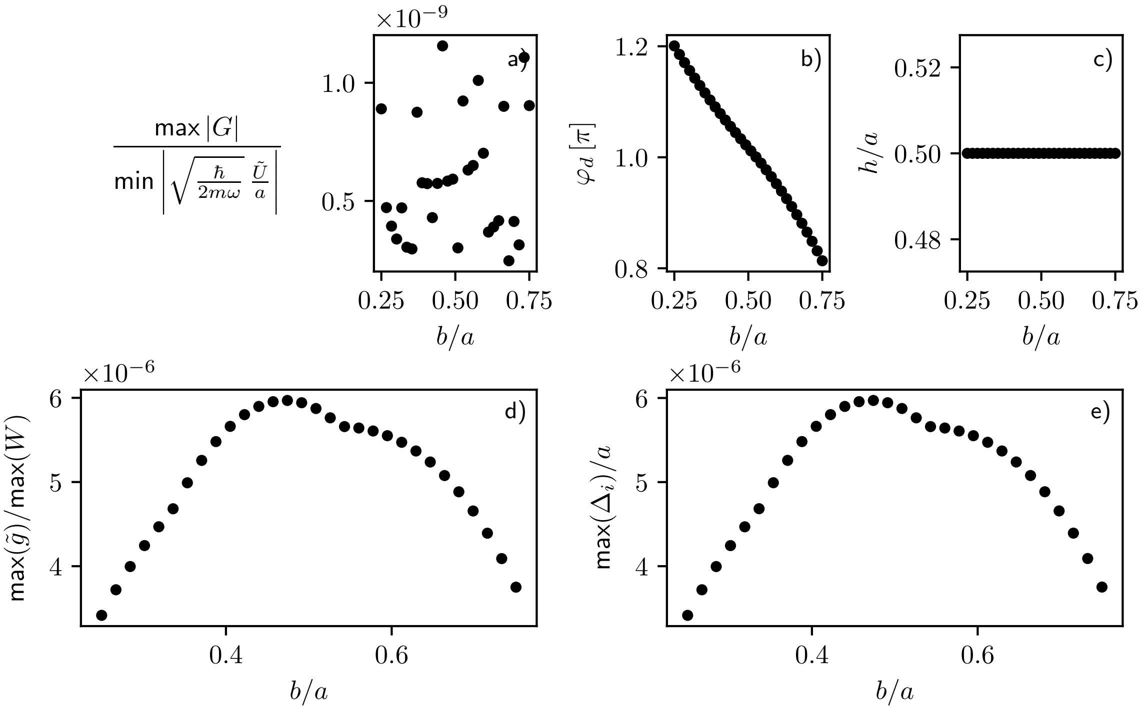

Figure 1: **Parameters for negligible first order.**

As shown in Fig. 1 the first order can be eliminated by choosing very small shifts in the displacement compared to the generic displacement inside the trap. This is possible due to the tuned small first order compared to the W-term as seen in Fig. 1d). Consequently the elimination of the linear term can also be understood from a comparison of the energy scales.

The Hamiltonian has now a quadratic form in the creation and annihilation operators and can hence be written as

$$
\begin{aligned}
H &= \sum_n \hbar\omega\, a_n^\dagger a_n + \frac{\hbar}{2m\omega} \sum_{n,m} U_{n,m} \left(a_n\, a_m + a_n^\dagger a_m + a_n\, a_m^\dagger + a_n^\dagger a_m^\dagger\right) \\
&= \mathbf{a}^\dagger A\, \mathbf{a}
\end{aligned}
$$

with $\vec{a} = (a_1, \ldots, a_N, a_1^\dagger, \ldots, a_N^\dagger)$ the vector of operators and a coupling matrix A. This allows for a topological characterization of the bosonic system.

Momentum space Hamiltonian

Let $c^A_{\mathbf{R},\alpha}$ and $c^B_{\mathbf{R},\alpha}$ be the phonon annihilation operators of sublattice A and B respectively at the unit cell, located at $\mathbf{R}$, where $\alpha \in \{x, y\}$ labels the the direction of movement. Let further L be the set of all unit cells of the lattice, and $C = \{c^A_{\mathbf{R},\alpha}, c^B_{\mathbf{R},\alpha}\}_{\mathbf{R}\in L, \alpha\in\{x,y\}}$ the set of all annihilation operators. The Hamiltonian in real space can then be written as

$$
H = \sum_{\substack{\mathbf{R},\mathbf{R}'\in L \\ a_\mathbf{R}\in C_\mathbf{R},\, b_{\mathbf{R}'}\in C_{\mathbf{R}'}}} U_{ab}(\mathbf{R}-\mathbf{R}') \left(a^\dagger_\mathbf{R} b_{\mathbf{R}'} + a_\mathbf{R} b^\dagger_{\mathbf{R}'} + a^\dagger_\mathbf{R} b^\dagger_{\mathbf{R}'} + a_\mathbf{R} b_{\mathbf{R}'}\right)
$$

For two dimensional periodic boundary conditions one can introduce the Fourier transformed operators, $a \in C$

$$
a_\mathbf{k} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}\in N} e^{-i\mathbf{k}\cdot\mathbf{R}}\, a_\mathbf{R},
$$

$k \in \bar{L}$ where $\bar{L}$ is the reciprocal lattice of L and N is the extent of the lattice in one direction. Writing the Hamiltonian in terms of these operators yields

$$
\begin{aligned}
H = \frac{1}{L} \sum_{\substack{\mathbf{R},\mathbf{R}' \in L \\ \mathbf{k},\mathbf{k}' \in \bar{L} \\ a_{\mathbf{k}} \in C_{\mathbf{k}},\, b_{\mathbf{k}'} \in C_{\mathbf{k}'}}} U_{ab}(\mathbf{R} - \mathbf{R}') \Bigg(& e^{-i\,(\mathbf{kR} - \mathbf{k}'\mathbf{R}')}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}'} + e^{i\,(\mathbf{kR} - \mathbf{k}'\mathbf{R}')}\, a_{\mathbf{k}} b^\dagger_{\mathbf{k}'} \\
& + e^{-i\,(\mathbf{kR} + \mathbf{k}'\mathbf{R}')}\, a^\dagger_{\mathbf{k}} b^\dagger_{\mathbf{k}'} + e^{i\,(\mathbf{kR} + \mathbf{k}'\mathbf{R}')}\, a_{\mathbf{k}} b_{\mathbf{k}'} \Bigg) \\
= \frac{1}{L} \sum_{\substack{\mathbf{R},\ \Delta\mathbf{R} = \mathbf{R} - \mathbf{R}' \\ \mathbf{k},\mathbf{k}' \\ a,b}} U_{ab}(\Delta\mathbf{R}) \Bigg(& e^{-i\,\mathbf{R}\,(\mathbf{k} - \mathbf{k}') - i\,\Delta\mathbf{R}\mathbf{k}'}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}'} + e^{i\,\mathbf{R}\,(\mathbf{k} - \mathbf{k}') + i\,\Delta\mathbf{R}\mathbf{k}'}\, a_{\mathbf{k}} b^\dagger_{\mathbf{k}'} \\
& + e^{-i\,\mathbf{R}\,(\mathbf{k} + \mathbf{k}') + i\,\Delta\mathbf{R}\mathbf{k}'}\, a^\dagger_{\mathbf{k}} b^\dagger_{\mathbf{k}'} + e^{i\,\mathbf{R}\,(\mathbf{k} + \mathbf{k}') - i\,\Delta\mathbf{R}\mathbf{k}'}\, a_{\mathbf{k}} b_{\mathbf{k}'} \Bigg) \\
= \sum_{\Delta\mathbf{R},\mathbf{k},a,b} U_{ab}(\Delta\mathbf{R}) & \left(e^{-i\,\Delta\mathbf{R}\mathbf{k}}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}} + e^{i\,\Delta\mathbf{R}\mathbf{k}}\, a_{\mathbf{k}} b^\dagger_{\mathbf{k}} + e^{-i\,\Delta\mathbf{R}\mathbf{k}}\, a^\dagger_{\mathbf{k}} b^\dagger_{-\mathbf{k}} + e^{i\,\Delta\mathbf{R}\mathbf{k}}\, a_{\mathbf{k}} b_{-\mathbf{k}} \right) \\
= \sum_{\mathbf{k},a,b} & \left(U_{ab}(\mathbf{k})\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}} + U^*_{ab}(\mathbf{k})\, a_{\mathbf{k}} b^\dagger_{\mathbf{k}} + U_{ab}(\mathbf{k})\, a^\dagger_{\mathbf{k}} b^\dagger_{-\mathbf{k}} + U^*_{ab}(\mathbf{k})\, a_{\mathbf{k}} b_{-\mathbf{k}} \right) =: \sum_{\mathbf{k}} H_{\mathbf{k}} \\
\text{where} \quad U_{ab}(\mathbf{k}) = \sum_{\Delta\mathbf{R}} & U_{ab}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{R}\mathbf{k}}
\end{aligned}
$$

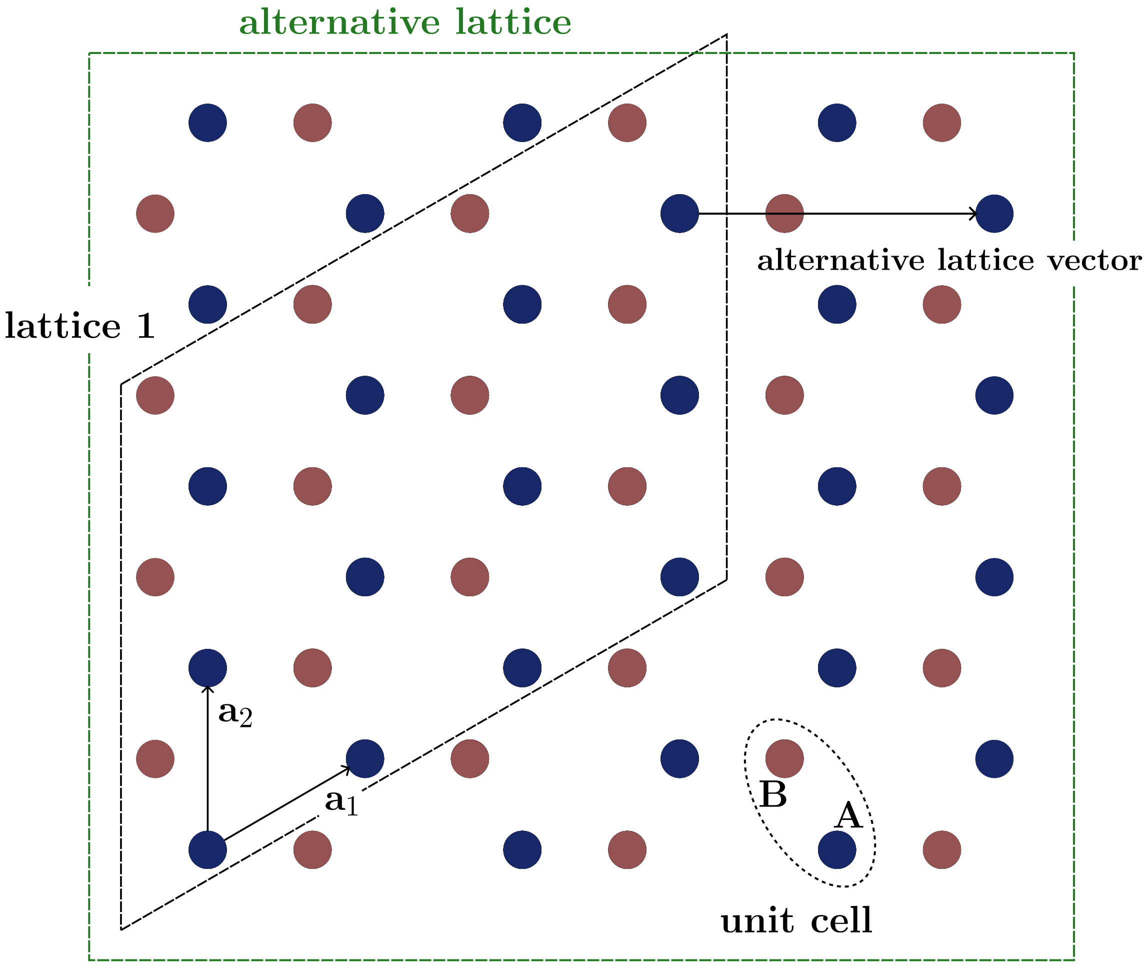

Figure 2: **Lattice definition** The main configuration of the grid,lattice 1, is the parallelogram spanned by the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. This is visualized by the black dashed frame. There is also the possibility of defining a square lattice which is highlighted by the green dashed frame.

For the case where only one direction of the lattice obeys periodic boundary conditions, while the other direction has open boundaries, the Fourier transform is only performed in the periodic direction. For this purpose I introduce the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$, which span the lattice. The lattice can be split into subspaces $L_n = \{j \cdot \mathbf{a}_n\}_{j \in M_n}$ whith $n \in \{1,2\}$ and M_n is chosen in such a way, that L_n spans from one edge of the lattice to the other. Hence the position of any unit cell $\mathbf{R}$ can be expressed as the linear combination of two vectors $\mathbf{r}_1 \in L_1$ and $\mathbf{r}_2 \in L_2$, i.e. $\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2$. I can now also introduce the reciprocal lattice vectors $\mathbf{k}_1$ and $\mathbf{k}_2$ with the property $\mathbf{a}_i \cdot \mathbf{k}_j = 2\pi\delta_{ij}$. I perform the Fourier transformation as an example for the lattice 1 from Fig. 2.

Here the lattice direction represented by $\mathbf{a}_2$ shall be periodic. I introduce the one dimensionally Fourier transformed operators

$$a_{\mathbf{r}_1,\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_2 \in L_2} e^{-i\mathbf{k}\mathbf{r}_2} \, a_{\mathbf{r}_1+\mathbf{r}_2}, \quad \mathbf{k} \in \{\frac{1}{N}, \ldots, 1\} \cdot \mathbf{k}_2$$

The Hamiltonian can now be transformed via

$$\begin{aligned}
H &= \frac{1}{N} \sum_{\substack{\mathbf{r}_1, \mathbf{r}_1' \in L_1 \\ \mathbf{r}_2, \mathbf{r}_2' \in L_2}} \sum_{\substack{a,b \in C \\ \mathbf{k}, \mathbf{k}' \in \bar{L}}} H^{a,b}(\mathbf{R} - \mathbf{R}')\, e^{-i\,\mathbf{k}\mathbf{r}_2}\, e^{i\,\mathbf{k}'\mathbf{r}_2'}\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}'} \\
&= \frac{1}{N} \sum_{\substack{\mathbf{r}_1, \mathbf{r}_1' \\ \mathbf{r}_2,\ \Delta\mathbf{r}_2 = \mathbf{r}_2 - \mathbf{r}_2'}} \sum_{\substack{a,b \\ \mathbf{k}, \mathbf{k}'}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\mathbf{R}(\mathbf{k}-\mathbf{k}')}\, e^{-i\,\Delta\mathbf{R}\mathbf{k}'}\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}'} \\
&= \sum_{\substack{\mathbf{r}_1, \mathbf{r}_1' \\ \Delta\mathbf{r}_2, \mathbf{k} \\ a,b}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{r}_2\mathbf{k}}\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}} =: \sum_{\substack{\mathbf{r}_1, \mathbf{r}_1' \\ \mathbf{k}, a, b}} H^{a,b}(\mathbf{k}, \Delta\mathbf{R})\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}}
\end{aligned}$$

$$\text{where} \quad H^{a,b}(\mathbf{k}, \Delta\mathbf{R}) = \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{r}_2\mathbf{k}} = \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{R}\mathbf{k}}$$

The last equality holds, when $\mathbf{k}$ is in the direction of one of the reciprocal lattice vectors. However, as we perform only a one dimensional Fourier transform, I think a restriction to a purely one dimensional treatment is necessary. Thus $\mathbf{k}_n$ should only be chosen with respect to the condition $\mathbf{k}_n \mathbf{a}_n = 2\pi$, which gives the natural choice $\mathbf{k}_n \parallel \mathbf{a}_n$. For the case of the alternative square lattice I think, we should consider the number of rows twice as big as the number of columns and choose an alternative lattice vector into the x-direction, as depicted in Fig. 2.

What the one dimensional Fourier transform does, is projecting the system onto the one-dimensional chain not of atoms, but of interacting rows or columns of atoms. In this sense the division into sub-directions or sub-bases does not have to be restricted to the lattice vectors. The only thing needed is a labeling of the different rows or columns with the complementary direction having a periodic structure. This allows the division of the alternative square lattice as described above.

Aligned Dipole-Dipole Interactions

In this section the derivative of the dipole interaction will be performed in the case where all dipoles point into the same direction. The subsripts lableling the different positions are droped in order to make the notation plainer.

$$V_{dd}(\mathbf{r}) = V_0 \left(\frac{1}{r^3} - 3\, \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^5} \right), \tag{0.1}$$

with $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j$ beeing the distance between any to sites i and j $(i \neq j)$. The first derivative with respect to component α of $\mathbf{r}$, $\alpha \in \{x, y, z\}$ reads

$$\frac{1}{V_0} \frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3 r_\alpha}{r^5} + \frac{15 r_\alpha (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - \frac{6 (\mathbf{r} \cdot \hat{\mathbf{d}}) d_\alpha}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{1}{V_0} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3\, \frac{\delta_{\alpha\beta}}{r^5} + 15\, \frac{r_\alpha r_\beta}{r^7} + 15\, \frac{\delta_{\alpha\beta} (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - 15 \cdot 7\, \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \\
&\quad + 30\, \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7}\, (\mathbf{r} \cdot \hat{\mathbf{d}}) - 6\, \frac{d_\alpha d_\beta}{r^5} \\
&= \delta_{\alpha\beta} \left(-3\, \frac{1}{r^5} + 15\, \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} \right) - 6\, \frac{d_\alpha d_\beta}{r^5} + 30\, \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7}\, (\mathbf{r} \cdot \hat{\mathbf{d}}) \\
&\quad + 15\, r_\alpha r_\beta \left(\frac{1}{r^7} - 7\, \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \right).
\end{aligned}$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$V_{dd}(\mathbf{r}) = V_0 \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3\, \frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^5} \right).$$

The first derivative reads

$$\frac{1}{V_0}\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)\, r_\alpha}{r^5} + \frac{15 r_\alpha (\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} - \frac{3\left((\mathbf{r}\cdot\hat{\mathbf{d}}_1)\, d_{2,\alpha} + (\mathbf{r}\cdot\hat{\mathbf{d}}_2)\, d_{1,\alpha}\right)}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{1}{V_0}\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3\, \frac{\delta_{\alpha\beta}\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)}{r^5} + 15\, \frac{r_\alpha r_\beta\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)}{r^7} + 15\, \frac{\delta_{\alpha\beta}(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} - 15\cdot 7\, \frac{r_\alpha r_\beta(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^9} \\
&\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7}\, (\mathbf{r}\cdot\hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7}\, (\mathbf{r}\cdot\hat{\mathbf{d}}_1) \right] - 3\, \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&= \delta_{\alpha\beta} \left(-3\, \frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^5} + 15\, \frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} \right) - 3\, \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7}\, (\mathbf{r}\cdot\hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7}\, (\mathbf{r}\cdot\hat{\mathbf{d}}_1) \right] \\
&\quad + 15\, r_\alpha r_\beta \left(\frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^7} - 7\, \frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^9} \right).
\end{aligned}$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$\begin{aligned}
\frac{\partial V_{dd}^2}{\partial r_\alpha} &= 2\, V_{dd} \frac{\partial V_{dd}}{\partial r_\alpha} \\
\frac{\partial^2 V_{dd}^2}{\partial r_\alpha \partial r_\beta} &= 2\, \frac{\partial V_{dd}}{\partial r_\alpha} \frac{\partial V_{dd}}{\partial r_\beta} + 2\, V_{dd} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta}
\end{aligned}$$

Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \tag{0.2}$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \tag{0.3}$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n}\cdot\boldsymbol{\sigma}, \tag{0.4}$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy-plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy-plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \to \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z}\, dz = \frac{1}{2\pi i} \int_{\mathrm{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \, \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads [6, 7]

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \, \text{Tr} \left[h^{-1}(k) \partial_k h(k) \right].$$

Diagonalizing h as $h = S\Lambda S^{-1}$ with the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, ...)$ and the invertible matrix S, the trace can be rewritten as

$$\begin{aligned}
\text{Tr} \left[h^{-1} \partial_k h \right] &= \text{Tr} \left[S\Lambda^{-1} S^{-1} \partial_k (S\Lambda S^{-1}) \right] \\
&= \text{Tr} \left[\Lambda^{-1} \partial_k \Lambda + S^{-1} \partial_k S + (\partial_k S^{-1}) S \right] \\
\text{Tr} \left[S^{-1} \partial_k S + S \partial_k S^{-1} \right] &= \text{Tr} \left[\partial_k \left(S S^{-1} \right) \right] = 0, \\
\Rightarrow \text{Tr} \left[h^{-1} \partial_k h \right] &= \text{Tr} \left[\Lambda^{-1} \partial_k \Lambda \right] = \sum_i \frac{\partial_k \lambda_i}{\lambda_i} = \partial_k \left(\sum_i \ln(\lambda_i) \right) = \partial_k \ln(\det[h]).
\end{aligned}$$

Another useful definition of the winding number is through the so-called Q-matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, ..., N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma |\psi_j\rangle = |\psi_{-j}\rangle$. The Q-matrix is defined as

$$Q = \sum_{j=1}^{N} \left(|\psi_j\rangle \langle \psi_j| - |\psi_{-j}\rangle \langle \psi_{-j}| \right).$$

As a consequence of the chiral symmetry, the Q-matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned}
\Gamma Q \Gamma &= \sum_{j=1}^{N} \left(|\psi_{-j}\rangle \langle \psi_{-j}| - |\psi_j\rangle \langle \psi_j| \right) = -Q \\
\Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}
\end{aligned}$$

The fact that the winding number of the q-matrix is the same as that of h can be shown with the help of the following two properties.

$$\begin{aligned}
Q^2 = 1 \quad &\Rightarrow \quad qq^\dagger = \mathbb{1}, \quad q^\dagger q = \mathbb{1} \\
QHQ = H \quad &\Rightarrow \quad qh^\dagger q = h^\dagger
\end{aligned}$$

From the second property follows that $q^\dagger h = h^\dagger q$. Thus $h^\dagger q$ is a hermitian matrix and therefore has no winding number, since the eigenvalues are real. Hence

$$\begin{aligned}
0 &= \text{Tr} \left[q^\dagger (h^\dagger)^{-1} \partial_k (h^\dagger q) \right] \\
&= \text{Tr} \left[q \, q^\dagger \, (h^\dagger)^{-1} \partial_k h^\dagger + h^\dagger (h^\dagger)^{-1} \, q^\dagger \partial_k q \right] \\
\Rightarrow \text{Tr} \left[q^{-1} \partial_k q \right] &= -\text{Tr} \left[(h^\dagger)^{-1} \partial_k h^\dagger \right] \\
&= \text{Tr} \left[h^{-1} \partial_k h \right]
\end{aligned}$$

There is another useful property, for which we consider the basis in which $\Gamma = \sigma_z \otimes \mathbb{1}_N$. This is the same diagonal basis as previously used. In this basis we can represent all eigenvalues of H as a stack of to N-component vectors. The eigenvector for negative energies is written as

$$|\psi_i^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |h_{1i}\rangle \\ |h_{2i}\rangle \end{pmatrix} \qquad i \in \{1, \dots, N\}$$

Because of the chirality of the Hamiltonian the eigenvector with positive energy is already determined by this choice and reads

$$|\psi_i^+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} |h_{1i}\rangle \\ -|h_{2i}\rangle \end{pmatrix}$$

The matrix Q is represented in the following form

$$\begin{aligned} Q &= \sum_i \begin{pmatrix} 0 & |h_{1i}\rangle\langle h_{2i}| \\ |h_{2i}\rangle\langle h_{1i}| & 0 \end{pmatrix} \\ \Rightarrow \quad q &= \sum_i |h_{1i}\rangle\langle h_{2i}| \end{aligned}$$

If we now compute the kernel of the winding number with respect to q we get

$$\begin{aligned} \mathrm{Tr}[q^\dagger \partial_k q] &= \mathrm{Tr}\left[\sum_{i,j} |h_{2i}\rangle\langle h_{1i}|\, \partial_k \left(|h_{1j}\rangle\langle h_{2j}|\right)\right] \\ &= \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle\langle h_{1i}| \left(\partial_k |h_{1j}\rangle\right) \langle h_{2j}|\right] + \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle\langle h_{1i}|\, |h_{1j}\rangle \left(\partial_k \langle h_{2j}|\right)\right] \\ &= \sum_i \langle h_{1i}| \left(\partial_k |h_{1i}\rangle\right) + \sum_i \left(\partial_k \langle h_{2i}|\right) |h_{2i}\rangle \end{aligned}$$

When computing the winding number the second term contributes with a minus sign compared to its complex conjugated self, i.e.

$$\int_{\mathrm{BZ}} dk\, \left(\partial_k \langle h_{2i}|\right)|h_{2i}\rangle = -\int_{\mathrm{BZ}} dk\, \langle h_{2i}| \left(\partial_k |h_{2i}\rangle\right)$$

Thus the winding number reads

$$\begin{aligned} w &= \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk\, \mathrm{Tr}[q^\dagger \partial_k q] \\ &= \frac{1}{2\pi i}\sum_i \int_{\mathrm{BZ}} dk\, \left(\langle h_{1i}|\partial_k|h_{1i}\rangle - \langle h_{2i}|\partial_k|h_{2i}\rangle\right) \end{aligned}$$

This can be expressed with respect to the full eigenvectors of the Hamiltonian via

$$\begin{aligned} w &= \frac{1}{\pi i}\sum_i \int_{\mathrm{BZ}} dk\, \langle \Gamma\psi_i^-|\,\partial_k|\psi_i^-\rangle \\ &= \frac{1}{\pi i}\sum_i \int_{\mathrm{BZ}} dk\, \langle \psi_i^+|\,\partial_k|\psi_i^-\rangle \\ &= \frac{1}{2\pi i}\sum_i \int_{\mathrm{BZ}} dk\, \left(\langle h_{1i}|\partial_k|h_{1i}\rangle - \langle h_{2i}|\partial_k|h_{2i}\rangle\right) \end{aligned}$$

The quantity $\sum_i \langle \Gamma\psi_i^-|\,\partial_k|\psi_i^-\rangle$ can be related to a polarisation [8].

Time-reversal symmetry

Time reversal symmetry is represented by an anti-unitary operator T. The effect of time-reversal on an operator is the following.

$$\hat{O} \to T\hat{O}T^{-1},$$

with $T^{-1} = T^\dagger$, as the norm of vectors has to be conserved. T can be decomposed into two operators $T = \tau K$, where K is the complex conjugation operator acting on everything to the right. T can satisfy one of conditions

$$T^2 = \pm 1$$

For spin-less particles in real space usually $\tau = \mathrm{id}$ and $T^2 = 1$, whereas for particles with spin $\tau = -i\sigma_y$ can be chosen in real space, which fulfills $T^2 = -1$.
Take a look at the momentum space Hamiltonian without spin degrees of freedom

$$
\begin{aligned}
H &= \sum_{\mathbf{k}} H(\mathbf{k})\, a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}} \\
\text{with} \quad a_{\mathbf{k}} &= \frac{1}{\sqrt{N^d}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\mathbf{R}}\, a_{\mathbf{R}}
\end{aligned}
$$

Time-reversal transforms the operators according to

$$
\begin{aligned}
T a_{\mathbf{R}} T^{-1} &= a_{\mathbf{R}} \\
\Rightarrow \quad T a_{\mathbf{k}} T^{-1} &= a_{-\mathbf{k}}
\end{aligned}
$$

Symmetry regarding time-reversal ($H = THT^{-1}$) has thus the form

$$
\begin{aligned}
H &= THT^{-1} \\
&= \sum_{\mathbf{k}} H^*(\mathbf{k})\, a_{-\mathbf{k}}^{\dagger} a_{-\mathbf{k}} \\
&= \sum_{\mathbf{k}} H^*(-\mathbf{k})\, a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}} \\
\iff \quad H(\mathbf{k}) &= H^*(-\mathbf{k})
\end{aligned}
$$

1D analysis of topological states

In order to have a chiral Hamiltonian I will choose the orientation of the dipoles in a way, such that the chirality remains. This will be done for each lattice configuration separately, i.e. the dipoles depend on the lattice geometry.
In order to make the achievement of chirality more feasible I only consider nearest neighbour interactions. When talking of chirality the chirality of the matrix U, defined above, is in mind. Recall that

$$
\begin{aligned}
U_{n,m} &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if}\, n \neq m \\
U_{n,n} &= \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}
\end{aligned}
$$

Chirality implies that all couplings between sites on the same sublattice vanish. The nearest neighbour restriction already assures that couplings between different sites of the same sublattice vanish. Left is thus only the onsite potential due to the taylor expansion of the dipole dipole potential. The aim is now to choose the orientaiton of the dipoles in such a way that the sum in $U_{n,n}$ is equal to zero.
Be aware that so far I didn't write explicitely the dependence of U from the direction of atom movement x, y. Hence $U_{n,m}$, $U_{n,n}$ are themselves two times two matrixes $(U_{n,m}^{\alpha,\beta})_{\alpha,\beta\in\{x,y\}}$ and the Hamilonian expressed explicitely with respect to the x, y-phonon modes reads

$$
H = \sum_{n,\alpha} \omega\, a_{n,\alpha}^{\dagger} a_{n,\alpha} + \frac{2}{m\omega} \sum_{n,m,\alpha,\beta} U_{n,m}^{\alpha,\beta}\, a_{n,\alpha}^{\dagger} a_{m,\beta}
$$

The terms with $a_{n,\alpha}^{\dagger} a_{n,\alpha}$ can thus be eliminiated by redefining the trap frequency.

$$
\omega_\alpha = \omega + \frac{2}{m\omega}\, U_{n,n}^{\alpha,\alpha}
$$

The terms coupling the two directions of phonon modes on the same side can not be internalized by another quantity. Here the degrees of freedom related to the dipole orientation are exploited to tune the $U_{n,n}^{x,y}$ values to zero, as mentioned above. Because of the nearest neighbour nature of the consideration only two distances play a role, the intracell distance $\mathbf{r}_I$ and the intercell distance $\mathbf{r}_O$.

$$
U_{n,n}^{x,y} = \frac{\partial^2 V(\mathbf{r}_I)}{\partial r_{I,x} \partial r_{I,y}} + \frac{\partial^2 V(\mathbf{r}_O)}{\partial r_{O,x} \partial r_{O,y}}
$$

The dipoles are then chosen in such a way that this sum vanishes. For open boundary conditions the values of $U_{n,n}^{\alpha,\beta}$ at the borders are different from the bulk values. In the case of $\alpha = \beta$ the difference can be compensated by changed trap frequencies at the boundary. For the xy coupling this is not possible and the interaction due to both neighboring atoms has to made zero individually such that x- and y-modes are completely decoupled from each other.

The topologically non-trivial phase actually breaks down, when those border-terms are not eliminated. This is confirmed by simulations. It might be, at first, counter intuitive that reducing the effect of an open border is decremental to the observation of edge states. It makes, however perfect sense, when remembering that those boundary terms in our case destroy the chiral symmetry. It seems to be necessary for the bulk-boundary condition to hold that not only the momentum space Hamiltonian with periodic boundary conditions has to satisfy chiral symmetry but also the real space Hamiltonian with open boundary conditions. Otherwise one could choose the Hamiltonian describing the border in a completely arbitrary manner and still observe edge states, what would be quite a strong statement.

What one observes is that, if the same direction edge coupling is kept in, no separated energies in the bulk gaps arise. At the other hand if only the boundary terms, coupling the two phonon directions, remain and $U_{n,n}^{\alpha,\alpha}$ is completely absorbed into the trap frequencies separated energies inside the bulk energy gaps can be observed. These separated energies, however, are not located at zero and converge for the atom number going to infinity to a non-zero value, what makes them non-topological.

Aligned dipoles in plane

I consider a geometry with two sublattices repeating itself in the x-axis and separeted by the distance h from each other into the y-axis. Different configurations of this geometry are realized by moving the sublattices against each other in the x-direction. I consider an even amount of lattice, with N atoms in each sublattice. The nth atom of each sublattice A and B takes the position.

$$\begin{aligned}\mathbf{r}_n^A &= n \cdot a\,\mathbf{e}_x\\ \mathbf{r}_n^B &= n \cdot a\,\mathbf{e}_x + b\,\mathbf{e}_x + h\,\mathbf{e}_y,\end{aligned}$$

where a is the lattice constant, and the parameters b and h can be chosen to realize different lattice configurations with the restriction that b and h have to be chosen in a way, such that the nearest neighbor restriction makes sense. For any b-value it is possible to find h and the dipole moment $\mathbf{d}$ in a way that fulfils the above conditions.

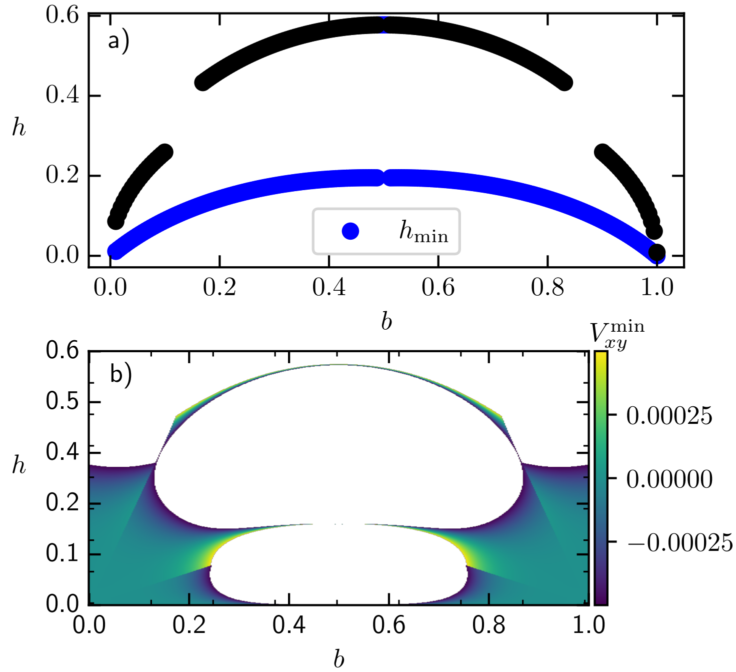

Figure 3: **Optimal h for chiral symmetry.** In **a)** The h values are depicted which decouple x- and y-phonons. Those values were determined numerically by computing roots of $V_x y(h)$. A grid of h values as initial guesses for the root algorithm was taken to have a better chance of finding all roots. In **b)** $V_x y(b, h)$ is drawn for reference. Larger values of the function have been omitted to make the pattern of small coupling values more visible.

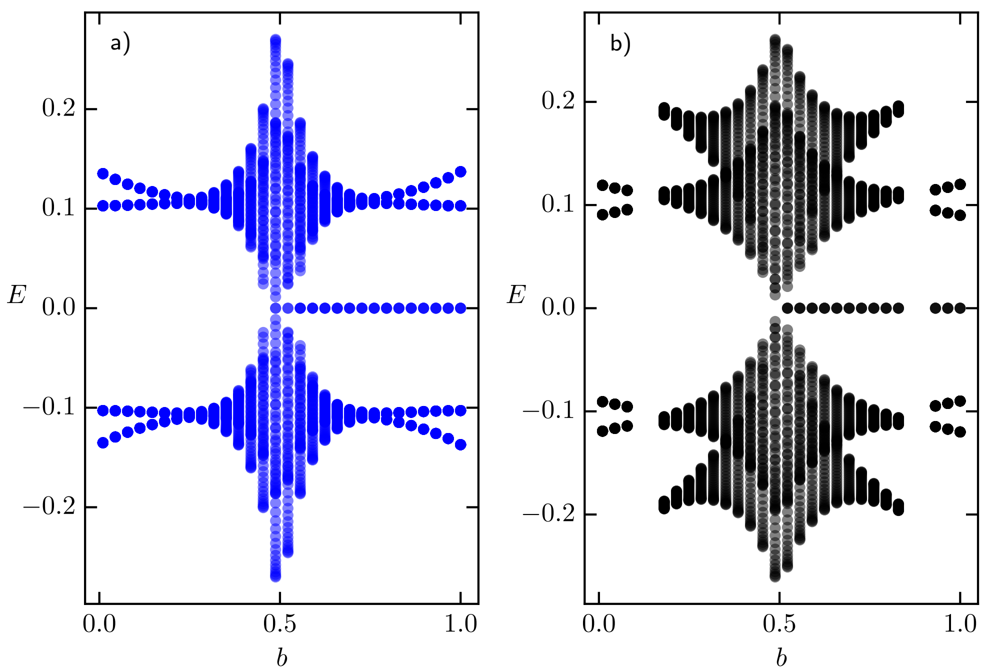

Figure 4: **Energy spectrum for the chiral Hamiltonian.**The diagram was created with the parameter $a = 1$. Blue and black colors depict the Energy spectrum for the h values picked from the blue and black curves in Fig. 3 respectively.

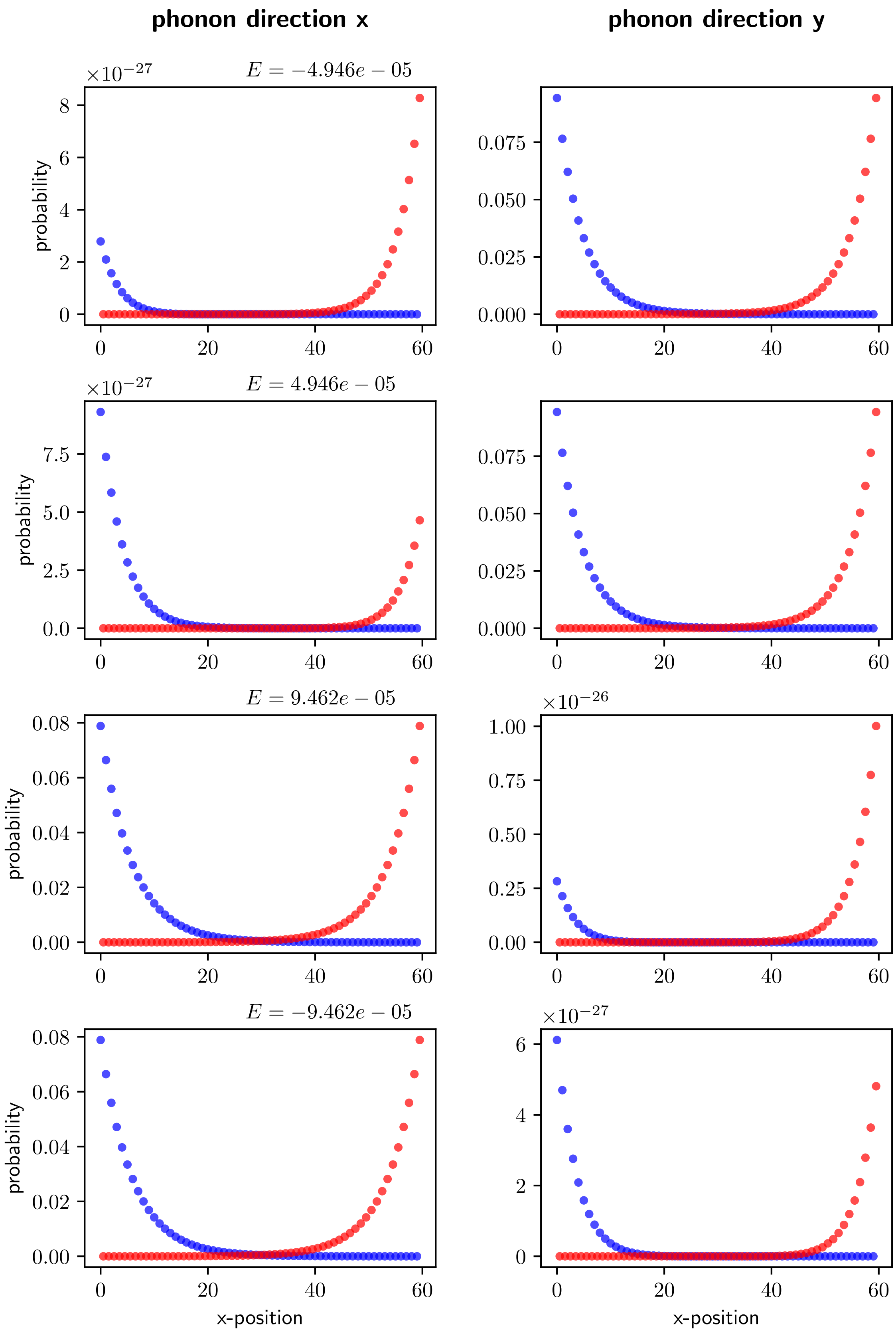

Figure 5: **Edge states**. The figure shows the decay of edge states with the four energies with the smallest absolute value. Each row depicts one energy state. Drawn in blue are the excitations of sublattice A, while the excitation of sublattice B is shown in red.

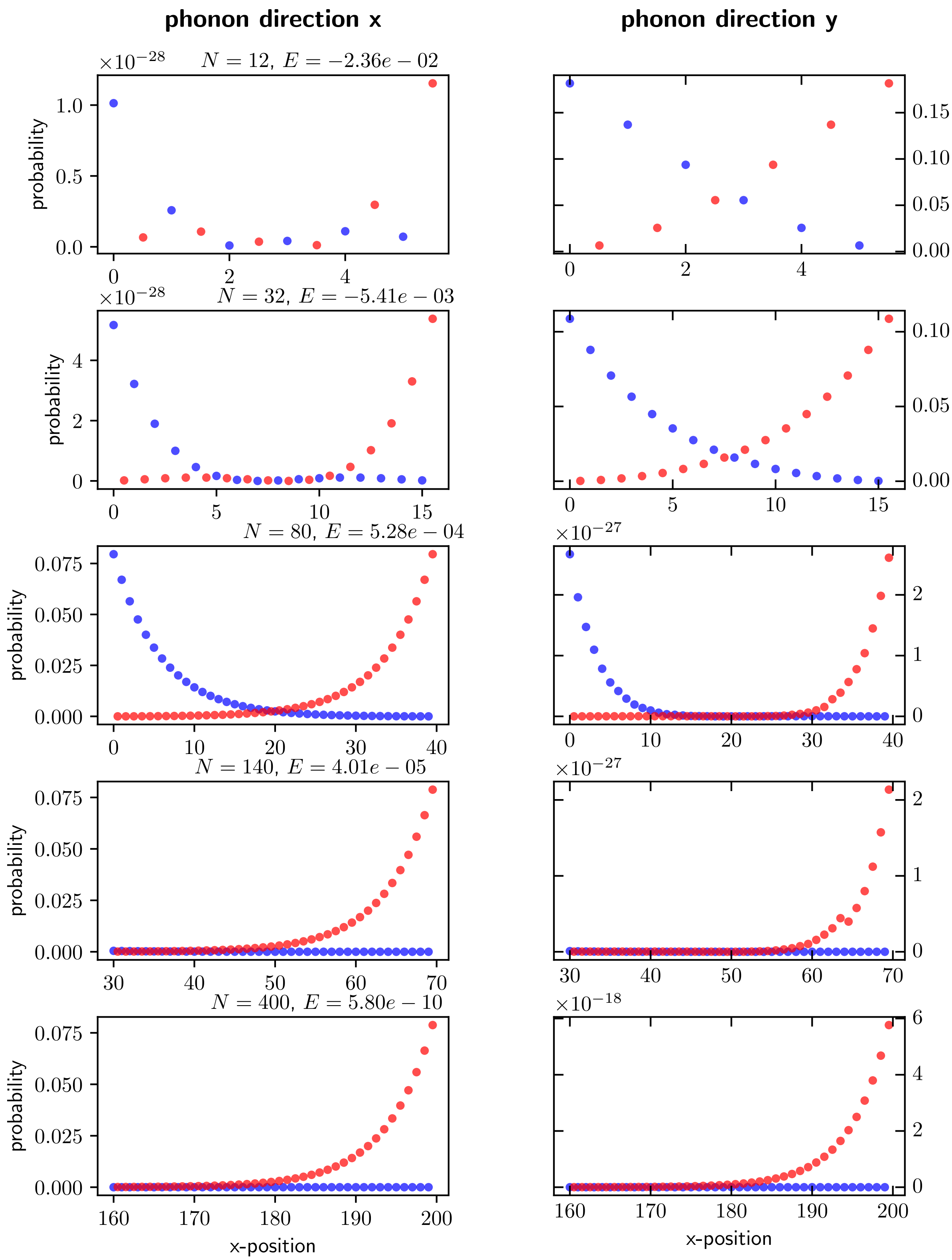

Figure 6: **Decay of energy for topological states with atom number.**. In each row the eigenstate for the energy with smallest absolute value is plotted. Blue and red show again the excitations of sublattice A and B respectively. When moving down in the rows of the figure the atom number of the system is increased. It shows that with increasing atom number the energy of the topological decays. What also can be observed. That the decay-length of the excitations when moving away from the edges does not change with particle number.

Spin-phonon coupling

Transport

I want to allow for the propagation of excitations through the chain. The first step towards this is constructing a Hamiltonian which describes the dynamics of a system of Rydberg atoms, where one atom is excited, hence restricting ourselves to the single excitation subspace. The general coupling between the atoms arises from the interaction between the dipole moments of the atoms,

$$V = \sum_{\mathbf{r}} V(\mathbf{r})$$
$$\text{with} \quad V(\mathbf{r}) = V_0 \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3\, \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5} \right).$$

Each individual atom is characterized by its position in the lattice and its atomic energy level which is captured by the non-interacting Hamiltonian H_0. The total Hamiltonian thus reads

$$H = H_0 + V$$

We will investigate the system in a regime, where the coupling due to the dipole forces is much smaller than the energy gap Δ, which separates the energy of the atoms from their next higher and lower states, i.e. $\|V\| \ll \Delta$. In order to gain an approximate effective Hamiltonian, which acts in a closed way on the one excitation subspace, I will perform a Schrieffer-Wolff transform up to second order [9]. Let P_0 be the projector on the one excitation subspace. Then an exact alternative Hamiltonian H' can be constructed by the transformation

$$H' = e^S\, H\, e^{-S}$$

for which holds, that it is block-diagonal with respect to the subspaces defined by the projections P_0 and $1 - P_0$. Thus the alternative Hamiltonian H' is closed with respect to the one excitation subspace and one can define the exact effective Hamiltonian as

$$H_{\text{eff}} = P_0\, e^S\, H\, e^{-S}\, P_0$$

Let ε be of the order of the coupling $\varepsilon = \|V\|$. The transformation generating operator S can be expanded in orders of V.

$$S = \sum_{n=1}^{\infty} \varepsilon^n\, S_n$$

I will consider the effective Hamiltonian up to second order in the coupling. This lets H_{eff} read

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \frac{1}{2}\, P_0\, [S_1, V_{\text{od}}]\, P_0,$$

with $V_{\text{od}} = P_0\, V\, (1 - P_0) + (1 - P_0)\, V\, P_0$ being the off-diagonal part of the interaction. The first term of the generator S has the form

$$S_1 = \mathcal{L}(V_{\text{od}})$$
$$\text{with} \quad \mathcal{L}(X) = \sum_{\substack{i,j \in E \\ E_i \neq E_j}} \frac{\langle i|V|j\rangle}{E_i - E_j}\, |i\rangle \langle j|$$

The sum in the superoperator $\mathcal{L}$ goes over all energy eigenstates, where the i, j combination of states with equal energy are omitted. It can be written via a sum over all energies in the single excitations subspace $E1$ and all the other eigenenergies $\tilde{E}$.

$$\mathcal{L}(X) = \sum_{\substack{i\in E1\\ j\in\tilde{E}}} \frac{\langle i|V|j\rangle}{E_i - E_j} \, |i\rangle\langle j| - \text{h.c.}$$

From this the commutator term in the effective Hamiltonian can be calculated.

$$P_0\left[S_1, V_{\text{od}}\right]P_0 = 2\sum_{\substack{i\in E1\\ j\in\tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} \, |i\rangle\langle i| =: H_{\text{eff},2}$$

Thus finally the effective Hamiltonian reads

$$H_{\text{eff}} = H_0P_0 + P_0VP_0 + \sum_{\substack{i\in E1\\ j\in\tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} \, |i\rangle\langle i|$$

For the effective Hamiltonian the contribution H_0P_0 is just a constant term and can be neglected. Further because of the large relative size of Δ over the interaction strength, I will only include energies into the above sum which differ from the single excitation subspace by Δ. Further the zero excitations subspace does not couple to the one with one excitation in the considered order of expansion. Thus only eigenstates in the double excitation subspace have to be taken into account for the sum.

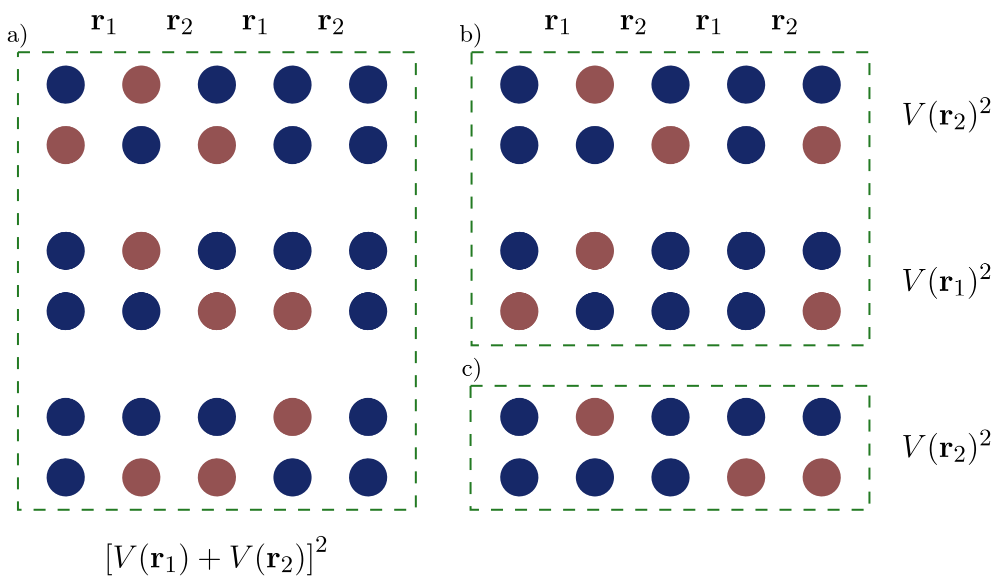

Figure 7: **Coupling schemes.** There are different ways double excited states couple to the single excited ones. The top row always depicts a state of the single, while the bottom row depicts a state of the double excited subspace. Generally configurations of two different subspaces are coupled to each other, when for two neighboring atoms one configuration has the inverted excitation scheme with respect to the other. I assign them to three categories. The first category is depicted in a). It couples two neighboring atoms twice. In the picture all possible cases are shown. For b) only two atoms couple two each other. Here the coupling takes place at a pair of $|e\rangle$ and $|g\rangle$ in both configurations. If the excitation in the single excitation subspace is at position $2 < i < N-1$, there are $N-4$ different states in the double excited subspace that contribute a term $V(\mathbf{r}_1)^2$ and the same number for $V(\mathbf{r}_2)^2$. Category c) couples two unexcited sites in the single excitations subspace to two excited sites in the double excited subspace. For this case there are $N/2-2$ states that contribute $V(\mathbf{r}_1)^2$ and $N/2-2$ states that contribute $V(\mathbf{r}_2)^2$, if the single excited state is is excited at atom $2 < i < N-1$.

In the following the effective Hamiltonian will be calculated for the specific configuration we are looking at. We consider a one-dimensional staggered chain of N atoms, where the atoms are separated by alternating distances $\mathbf{r}_1$ and $\mathbf{r}_2$. For a single atom there are only two states available, excited $|e\rangle$ and unexcited $|g\rangle$. The states $|i\rangle$, which are summed over, represent the states where only atom i is excited and the others are in the unexcited state. Going to a regime where only nearest neighbor interactions contribute we can write using the coupling schemes of Fig. 7

$$H_{\text{eff},2} = \sum_{i=1}^{N} \tilde{V}_i \; |i\rangle \langle i|$$

$$\tilde{V}_i = -\frac{1}{\Delta} \begin{cases} 3\,[V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N - 4 + \frac{N}{2} - 2)\, V^2(\mathbf{r}_1) + (N - 4 + \frac{N}{2} - 2)\, V^2(\mathbf{r}_2) & \text{if } 1 \neq i \neq N \\ [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N - 3 + \frac{N}{2} - 1)\, V^2(\mathbf{r}_1) + (\frac{N}{2} - 1)\, V^2(\mathbf{r}_2) & \text{if } i = 1 \\ 2\,[V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N - 3 + \frac{N}{2} - 2)\, V^2(\mathbf{r}_1) + (N - 4 + \frac{N}{2} - 1)\, V^2(\mathbf{r}_2) & \text{if } i = 2 \\ 2\,[V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (N - 4 + \frac{N}{2} - 1)\, V^2(\mathbf{r}_1) + (N - 3 + \frac{N}{2} - 2)\, V^2(\mathbf{r}_2) & \text{if } i = N-1 \\ [V(\mathbf{r}_1) + V(\mathbf{r}_2)]^2 + (\frac{N}{2} - 1)\, V^2(\mathbf{r}_1) + (N - 3 + \frac{N}{2} - 1)\, V^2(\mathbf{r}_2) & \text{if } i = N \end{cases}$$

The part involving interactions within the same subspace can be written as

$$P_0 V P_0 = \sum_{\substack{i=1 \\ j=i+1}}^{N-1} V(\mathbf{r}_{i,j})\, \sigma_i^+ \sigma_j^- + \text{h.c.},$$

where $\mathbf{r}_{i,j}$ is the distance between the ith and jth site. I now also want to write $H_{\text{eff},2}$ in terms of a sum of two atom operators. Therefore I introduce the projectors $n_i^e = |e\rangle_i \langle e|_i$ and $n_i^g = |g\rangle_i \langle g|_i$.

$$\begin{aligned} H_{\text{eff},2} = & \sum_{\substack{i=1 \\ j=i+1}}^{N-1} V^2(\mathbf{r}_{i,j})\,(n_i^e n_j^g + n_i^g n_j^e + n_i^g n_j^g) + \sum_{i=2}^{N-1} [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 \; n_{i-1}^g n_i^e n_{i+1}^g \\ & + \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 \; n_{i-2}^g n_{i-1}^g n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 \; n_i^e n_{i+1}^g n_{i+2}^g \\ & + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2\, n_1^e n_2^g n_3^g + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2\, n_{N-2}^g n_{N-1}^g n_N^e \end{aligned}$$

A last step is possible since in the single excitation subspace terms like $n_i^e n_j^g$ simplify to just n_i^e.

$$\begin{aligned} = & \sum_{i=2}^{N-1} \Big(V^2(\mathbf{r}_{i-1,i}) + V^2(\mathbf{r}_{i,i+1}) + [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 \Big) \; n_i^e \\ & + \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 \; n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 \; n_i^e \\ & + \Big(V^2(\mathbf{r}_{1,2}) + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2 \Big)\; n_1^e + \Big(V^2(\mathbf{r}_{N-1,N}) + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2 \Big)\; n_N^e \\ & + \sum_{i=1}^{N-1} V(\mathbf{r}_{i,i+1})\, n_i^g n_{i+1}^g \end{aligned}$$

Energy scales

The most important energies which determine the dynamics are the trap frequency, which is of the order $\omega \sim 10$ - $100\,$kHz, the distances between atoms who can reliably reach $3\,\mu$m, and the lifetime of the Rydberg states, which is in the order of $100\,\mu$s. From this it already becomes apparent that, in order to see dynamics of the system within the lifetime of the Rydberg states, the coupling strength has to be higher than the trap frequency. This prevents the Lamb-Dicke regime, as the phonon number cannot be assumed as conserved. Thus the first order expansion in the deplacements of the atoms has to be taken into account. The relative size of each order of the expansion can be estimated by the factor

$$\varepsilon := \sqrt{\frac{\hbar}{m\omega}}\frac{1}{r} \approx (2\text{-}11)\cdot 10^{-3}$$

As the dipole interaction is a function of the angle ϑ between the dipole moment and the axis of the chain, the interaction can be tuned in a way that the first order term vanishes. As can be seen in Fig. 8, the second order term in the expansion is of the order $5\cdot 10^{-5}\text{-}10^{-3}$, smaller than the leading order.

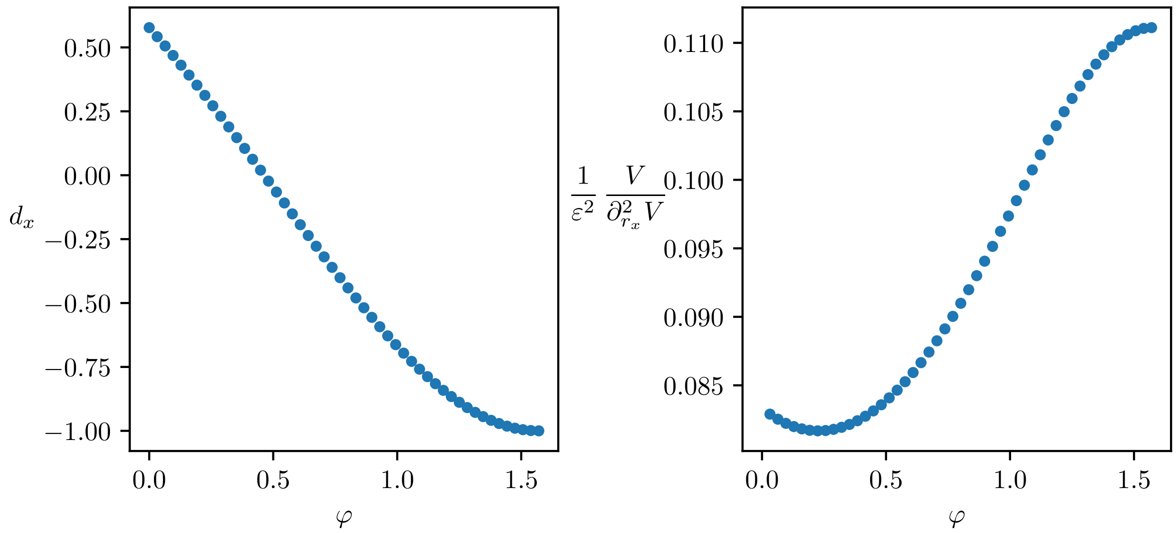

Figure 8: **Dipole orientation and ratio of interactions.** For all angles $\sin\varphi := h/r$ one can find a dipole orientation, shown in the left panel, such thatthe first order term in the expansion of the interaction vanishes. The ratio between zeroth order and second order term is shown in the right panel.

Spin-Phonon Hamiltonian

Let us first clarify the approximations and choices made that lead to the spin-phonon coupled Hamiltonian that is used in the following. The consideration is restricted to the single-excitation subspace with respect to the electronic degrees of freedom, i.e the spin degrees of freedom. This is achieved by going into a regime where the energy distance between states is much larger than their coupling. As a second feature the system shall be arranged in a way that the number of motional excitations is conserved as well. This is achieved by tuning the linear phonon creation term, stemming from the first order Taylor expansion of the coupling potential, to zero. This procedure is laid out in the following.

Considering only nearest neighbor interactions the interaction Hamiltonian expanded to the second order in the displacements from the trap center reads

$$\begin{aligned} H_{\text{int}} &= \sum_{i,j=1;\, i<j}^{N-1}\left[V(\mathbf{r}_{i,j}) + \sum_{n\in\{i,j\}}\frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{R}_n}\,\delta\hat{\mathbf{R}}_n + \frac{1}{2}\sum_{n,m\in\{i,j\}}\frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{R}_n\partial \mathbf{R}_m}\,\delta\hat{\mathbf{R}}_n\delta\hat{\mathbf{R}}_m\right]\sigma_i^+\sigma_j^- + \text{h.c.} \\ &= \sum_{i,j=1;\, i<j}^{N-1}\left[V(\mathbf{r}_{i,j}) + \frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}}\,(\delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_i)\right. \\ &\qquad\left. + \frac{1}{2}\frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2}\left(\delta\hat{\mathbf{R}}_i^2 + \delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_i\,\delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_j\delta\hat{\mathbf{R}}_i\right)\right]\sigma_i^+\sigma_j^- + \text{h.c.} \end{aligned}$$

As this step one can introduce a shift in the displacement as was done in Section ,i.e. $\{\delta\mathbf{R}_l\} \to \{\delta\mathbf{R}_l + \mathbf{\Delta}_l\}$. Plugging the shifted displacements in the above equation and considering only the terms linear in displacement (H_{Lin}), one can collect the following terms.

$$
\begin{aligned}
H_{\text{Lin}} = \sum_{i,j=1;\, i<j}^{N-1} \Bigg[&\left(-\frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}} - \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} (\Delta_j - \Delta_i) \right) \delta\hat{\mathbf{R}}_i \\
&+ \left(\frac{\partial V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}} + \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} (\Delta_j - \Delta_i) \right) \delta\hat{\mathbf{R}}_j \Bigg] \sigma_i^+ \sigma_j^- + \text{h.c.} \\
= \sum_{i,j=1;\, i<j}^{N-1} &\left[\frac{V_0}{a} \left(\frac{a^4}{r_{i,j}^4}\, g(\hat{\mathbf{r}}_{i,j}) + \frac{a^5}{r_{i,j}^5}\, c(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right) \right) (\delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_i) \right] \sigma_i^+ \sigma_j^- + \text{h.c.}
\end{aligned}
$$

This constitutes the condition which the shifts in displacement have to satisfy.

$$
G_{i,j} := \frac{V_0}{a} \left(\frac{a^4}{r_{i,j}^4}\, g(\hat{\mathbf{r}}_{i,j}) + \frac{a^5}{r_{i,j}^5}\, c(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right) \right) = 0 \quad \forall i,j \in \{1, N-1\};\ i<j
$$

With this definition one can write the interaction Hamiltonian as

$$
\begin{aligned}
H_{\text{int}} = \sum_{i,j=1;\, i<j}^{N-1} \Bigg[&V(\mathbf{r}_{i,j}) + \Delta V_{i,j} + G_{i,j}\, (\delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_i) \\
&+ \frac{1}{2} \frac{\partial^2 V(\mathbf{r}_{i,j})}{\partial \mathbf{r}_{i,j}^2} \left(\delta\hat{\mathbf{R}}_i^2 + \delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_i\, \delta\hat{\mathbf{R}}_j - \delta\hat{\mathbf{R}}_j \delta\hat{\mathbf{R}}_i \right) \Bigg] \sigma_i^+ \sigma_j^- + \text{h.c.} \\
\text{with} \quad \Delta V_{i,j} = V_0\, &\frac{a^4}{r_{i,j}^4}\, g(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right) + \frac{1}{2} V_0\, \frac{a^5}{r_{i,j}^5}\, c(\hat{\mathbf{r}}_{i,j}) \left(\frac{\Delta_j}{a} - \frac{\Delta_i}{a} \right)^2
\end{aligned}
$$

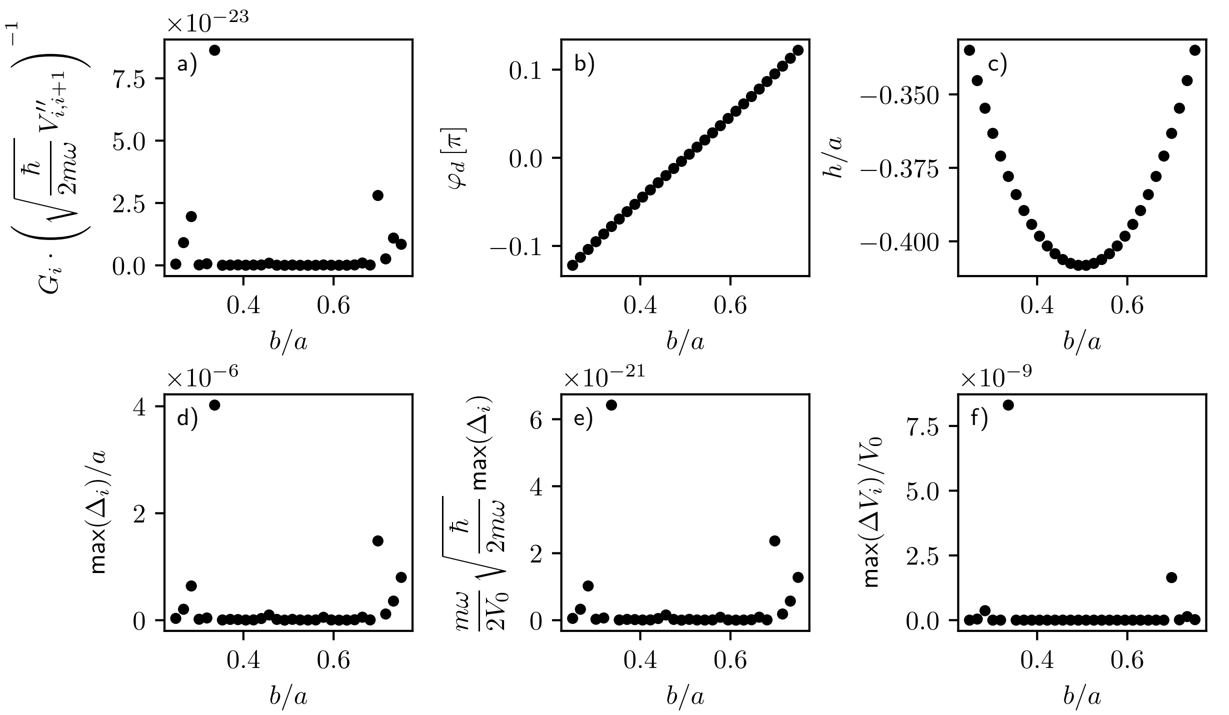

Figure 9: **Parameters for negligible first order.**

First order approximation in spin degrees

For the first approach I will continue with the Hamiltonian up to first order in the dipole interaction, i.e. without $H_{\text{eff},2}$. The phonons are introduced by again expanding the interaction up to second order.

When we choose the dipole orientation, such that thr first order vanishes, the Hamiltonian reads

$$
\begin{aligned}
H &= \sum_{i=1}^{N-1} \left[V(\mathbf{r}_{i,i+1}) + \frac{1}{2} \sum_{n,m\in\{i,i+1\}} \frac{\partial^2 V(\mathbf{r}_{i,i+1})}{\partial \mathbf{r}_n \partial \mathbf{r}_m} \delta\hat{\mathbf{r}}_n \delta\hat{\mathbf{r}}_m \right] \sigma_i^+ \sigma_{i+1}^- + \text{h.c.} \\
&= \sum_{i=1}^{N-1} \left[V(\mathbf{r}_{i,i+1}) - \frac{\hbar}{2m\omega} \frac{\partial^2 V(\mathbf{r}_{i,i+1})}{\partial \mathbf{r}_i^2} \left(p_j^\dagger p_{j+1} + p_{j+1}^\dagger p_j - p_j^\dagger p_j - p_{j+1}^\dagger p_{j+1} \right) \right] \sigma_i^+ \sigma_{i+1}^- + \text{h.c.}
\end{aligned}
$$

The quadrature operators of the phonons are used $\delta\hat{\mathbf{r}}_i = \sqrt{\frac{\hbar}{2m\omega}}\,(p_i + p_i^\dagger)$. In the following I want to present a brief investigation of the influence of this small second order term on the dynamics of the spins. Ultimately I am interested in the effect of the phonons on the transport of spin excitations via a Rice-Mele scheme. Therefore I will compare the free evolution of the system with and without phonons at conditions mimicking a point in the middle of the Rice-Mele cycle, where the dynamics are the fastest. The results are shown in Fig. 10. As can be seen, for high phonon excitation numbers the difference in the spin excitations during the time period of the whole cycle becomes substantial. Nevertheless, the performance of the Rice-Mele protocol is affected only mildly, and is even slightly improved by the inclusion of the phonons, as can be seen in Fig. 13.

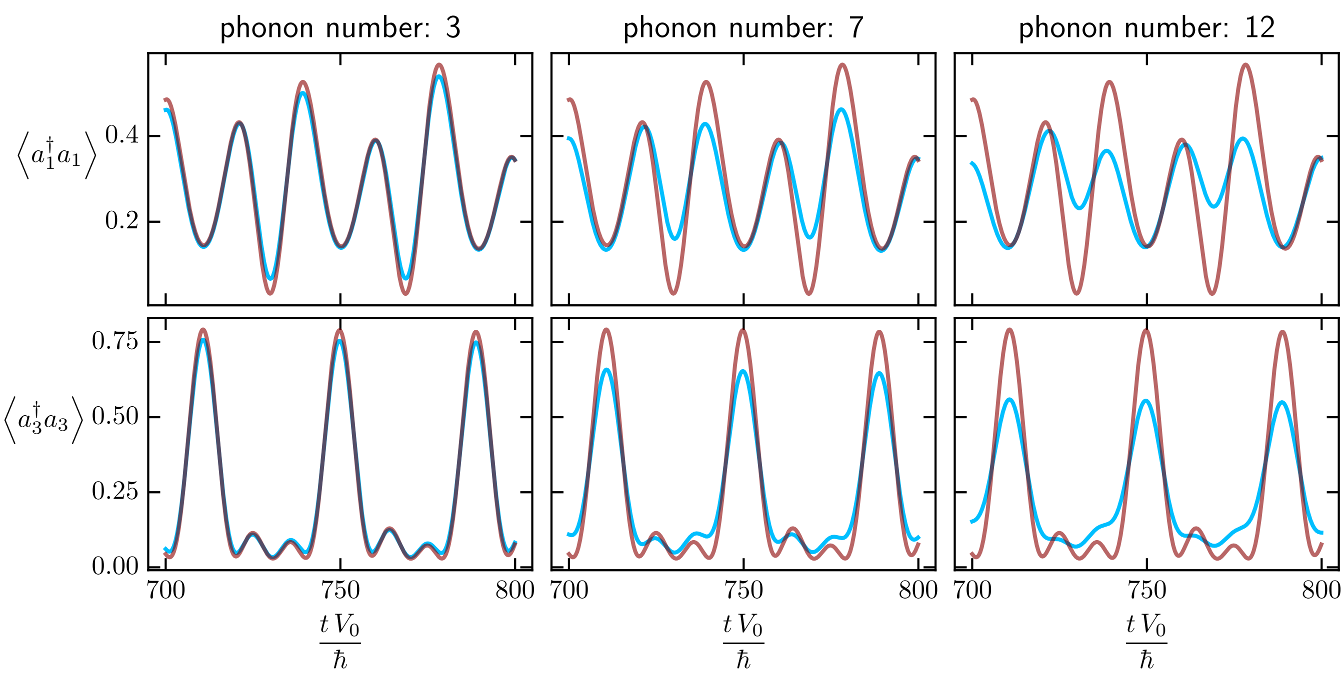

Figure 10: **Comparison of free evolution with and without phonons.** The difference in spin excitation probability between the free evolution with and without phonons is shown for a chain of length $N = 4$. As initial conditions I chose a quadratically decaying excitation profile from the right edge and parameters $b = 1.2\,r_0$, $h = 0.4\,r_0$, $a = 3\,r_0$. The red line shows the trajectory for the case without motional degrees of freedom and the blue one the case with phonons present.For an atom with 12 motional excitations the mean displacement is of the order of $0.15\,\mu$m, which is already around the limit of the expansion to second order being accurate.

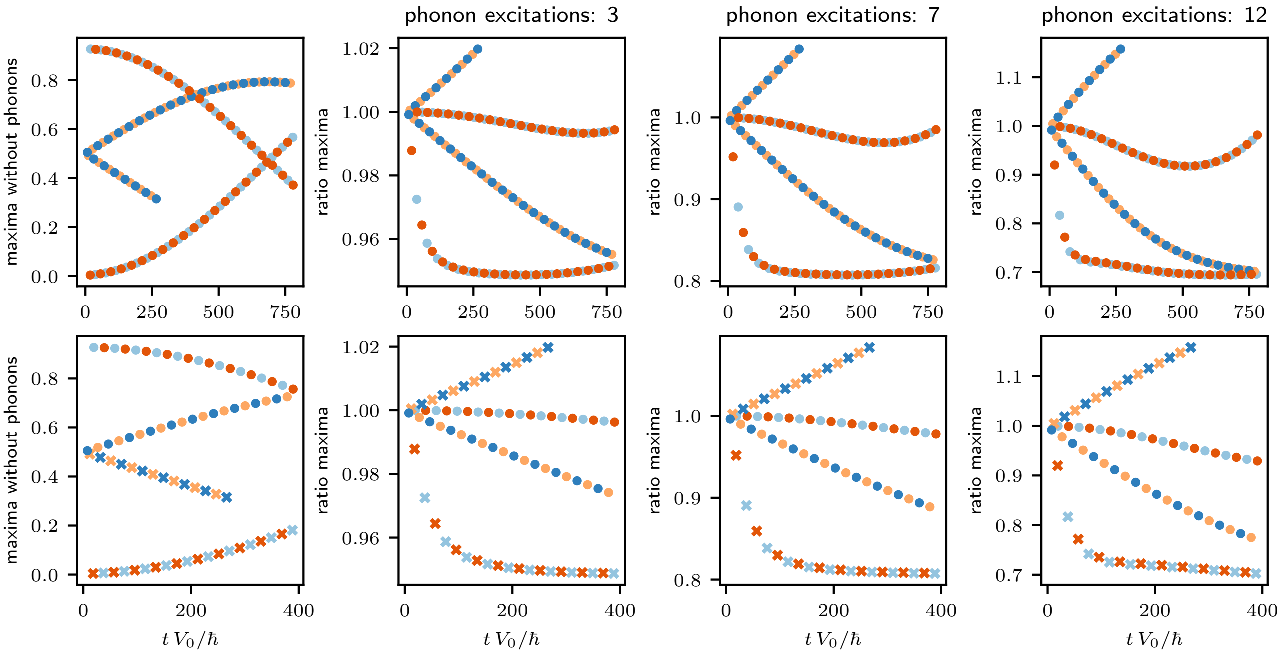

Figure 11: **Comparison of the maxima of the free evolution with and without phonons.** For the same parameters and initial conditions as in Fig. 10 the maxima of the spin excitation probability at the individual sites are compared for the free evolution with and without phonons.

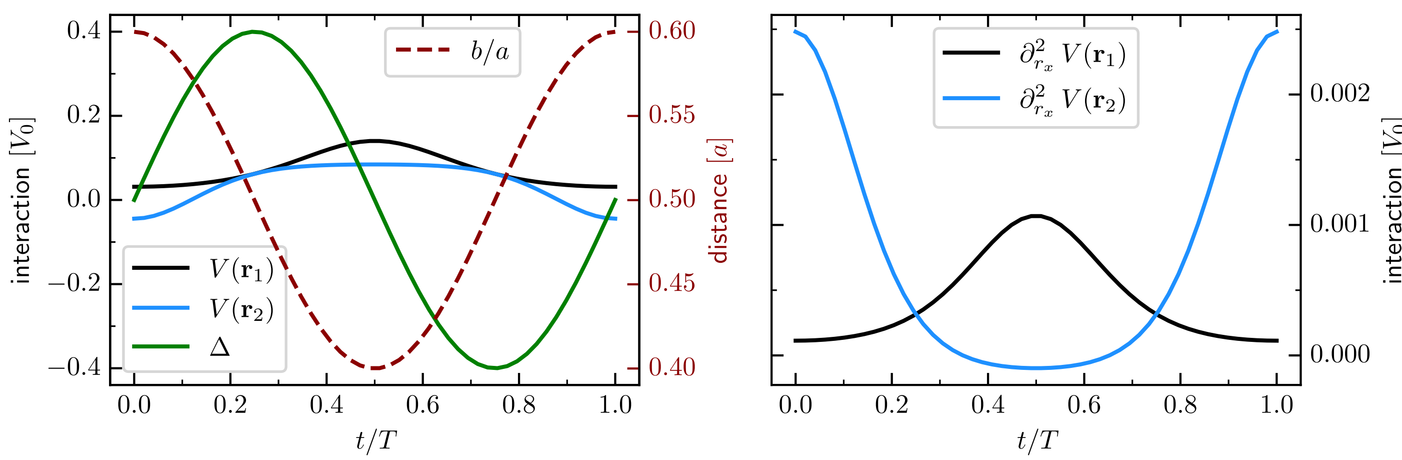

Figure 12: **Rice-Mele scheme.** Depicted is the course of the the decisive parameters during the Rice-Mele cycle, i.e. the axial distance between atoms in the unit cell $b = 1.8 \text{-} 1.2 \text{-} 1.8\, r_0$ (the distance orthogonal to the chain axis is kept constant $h = 0.4\, r_0$ as well as the distance inside a sublattice $a = 3\, r_0$). The detuning takes it's maximal value at $\Delta_{\max} = 0.4\, V_0$.

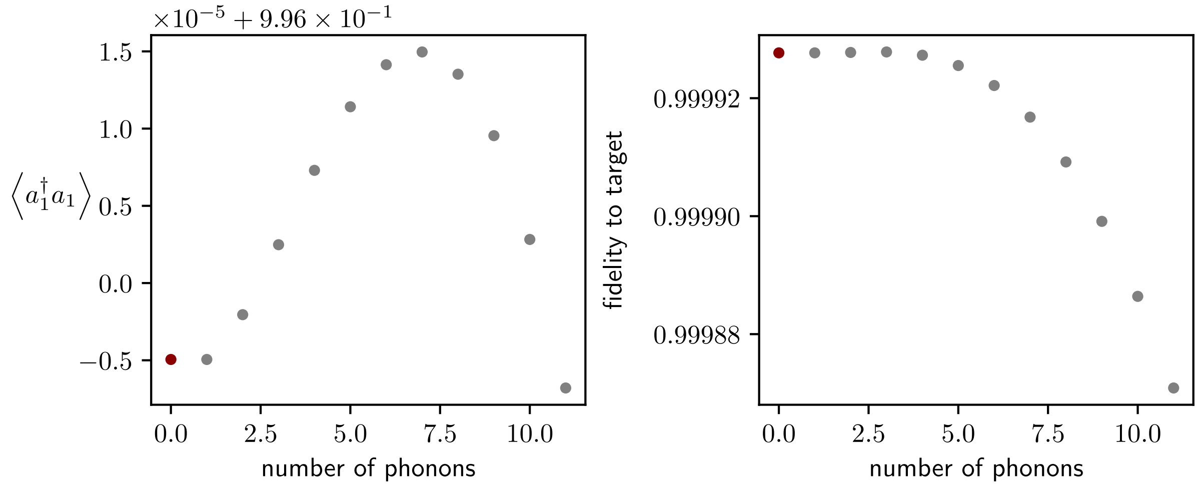

Figure 13: **Transfered excitation in dependence of phonon number.** I show the excitation probability at the left edge at the end of the cycle, when the system is initialized in an eigenstate of the pure spin Hamiltonian localized at the right edge. And parameters of the cycle are $b = 1.8\text{-}1.2\text{-}1.8\,r_0$, $h = 0.4\,r_0$, $a = 3\,r_0$, $\Delta_{\max} = 0.4\,V_0$.

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